

Fall 25, 4100 Problem Set 9

Please complete all problems. For any proof questions, please structure your proof similar to those from lecture.

1. Sketch the conic section $x_1^2 + 8x_1x_2 + 7x_2^2 = 1$ by finding the spectral decomposition of the matrix $A = \begin{bmatrix} 1 & 4 \\ 4 & 7 \end{bmatrix}$
2. Let A be any real matrix and prove that
 - a. If $\mathbf{v}$ is an eigenvector of AA^T , then $\mathbf{v} \in \text{range}(A) \cup \text{null}(A^T)$ (look at the cases where the corresponding e-value is 0 or nonzero)
 - b. If λ is a nonzero eigenvalue of AA^T , then it is also an eigenvalue of A^TA

3. Let

$$A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \\ 1 & 2 \end{bmatrix}$$

- a. Find the singular value decomposition of A
 - b. Find the psuedo-inverse, A^+ and use it to find the vector in range A that is closest to $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$
4. Find and sketch the image of the unit circle under the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ where

$$\begin{aligned} T\mathbf{e}_1 &= 2\mathbf{e}_1 + \mathbf{e}_2 \\ T\mathbf{e}_2 &= -3\mathbf{e}_1 + 6\mathbf{e}_2 \end{aligned}$$

using the method from lecture 11-20

5. Given $n \times n$ matrix A , define the trace of A as

$$\text{tr}(A) = \sum_{i=1}^n A_{i,i} = \langle \mathbf{e}_1, A\mathbf{e}_1 \rangle + \cdots + \langle \mathbf{e}_n, A\mathbf{e}_n \rangle$$

Now letting $A \in \mathbb{C}^{n \times n}$ have singular values of $\sigma_1, \dots, \sigma_r$, show that

$$\text{tr}(A^*A) = \sigma_1^2 + \cdots + \sigma_r^2$$